

AI WeatherWise

AI-Augmented Backend Development Project

PHASE 3 – PROJECT DESIGN

Team No.	05
Team ID	SWTID-2026-9139
Project Title	AI WeatherWise
Team Leader	HIBA FATHIMA S
Team Members	DIVYA SREE V, LINCYA A H, RESHMA J

1. Project Design Overview

The Project Design phase transformed the approved requirements into a structured backend architecture for AI WeatherWise. The design focuses on modular development, secure authentication, database relationships, API communication, and scalable request handling.

Design Objective

Create a clean backend architecture where users can authenticate securely, access weather services, save favourite locations, and receive AI-powered recommendations through REST APIs.

2. System Architecture

The backend follows a layered architecture that separates routing, controllers, services, middleware, and database models for better maintainability.

Client → Express Server → Authentication Middleware → Controllers → Services → MongoDB / Weather API / AI Recommendation Service

Architecture Diagram

Insert the actual System Architecture diagram from your Main Documentation in this section while editing the document.

3. Database Design

MongoDB stores user accounts and favourite locations while maintaining a one-to-many relationship between users and locations.

Collection	Fields
User	id, name, email, password, role
Location	id, user, city, country, createdAt
Relationship	One User → Many Locations
Authentication	JWT Token
Security	Hashed Passwords

4. Backend Folder Structure

The backend is organised into modular folders to simplify development and maintenance.

Folder Structure
src/ config controllers middleware models routes services app.js server.js

5. API Design

The REST API design enables secure communication between clients and backend services.

API	Method	Purpose
Register	POST	Create user account
Login	POST	Authenticate user
Weather	GET	Retrieve weather
Recommendation	POST	Generate weather recommendation

6. Design Decisions

- Node.js and Express.js for backend development.
- MongoDB with Mongoose for flexible data storage.
- JWT authentication for secure access.
- Centralized error handling for consistent responses.
- Modular architecture for future scalability.

7. Design Outcome

The Project Design phase produced a structured backend blueprint that supports secure authentication, weather retrieval, MongoDB persistence, AI recommendations, and scalable REST API development. This design became the implementation guide for the Project Development phase.